z-transforms and linear recursions

1. Signal Processing
   - sampling a signal
   - capturing growth or decay
   - delayed signals

2. Linear Recursions
   - solving recursions via the z-transform
   - partial fraction decomposition
   - using `sympy.rsolve`

MCS 472 Lecture 6
Industrial Math & Computation
Jan Verschelde, 26 January 2026

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a motivating problem

A train leaves the depot empty.

At each station, on average,

- 40 people mount,
- while 20% of the riders dismount.

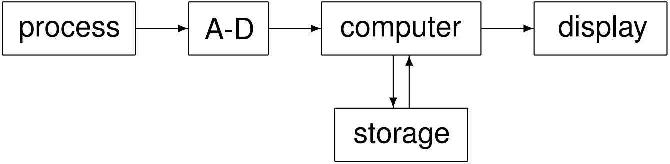
Suppose we are consultants for a train company, who asks us:

Will there ever be a train long enough

to accommodate the 40 people mounting at each station?

acquiring and processing data

A schematic for a *data acquisition system*:

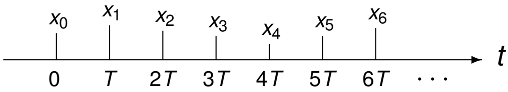


1. A process is measured by a sensor.
2. The measurements are converted from analog to digital (A-D).
3. The computer manipulates, stores, and displays the data.

We are concerned with the third stage of the acquisition process.

the z-transform of a signal

Every T time units (seconds or days), we sample a signal:



The samples of the signal define a sequence x in the *time domain*:

$$x = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, \ldots\}.$$

In the *frequency domain*, we define the Laurent series

$$X = \sum_{k=0}^{\infty} x_k z^{-k} = x_0 + \frac{x_1}{z} + \frac{x_2}{z^2} + \frac{x_3}{z^3} + \frac{x_4}{z^4} + \frac{x_5}{z^5} + \frac{x_6}{z^6} + \cdots$$

as *the z-transform of the signal*, as the series converges for $|z| > 1$.

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a constant signal

In the series

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + \cdots$$

replace z by $1/z$:

$$\frac{1}{1-z^{-1}} = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} + \cdots$$

So the z-transform of $x = \{1, 1, 1, 1, 1, 1, 1, \ldots\}$ is

$$X = \frac{1}{1-z^{-1}} = \frac{z}{z-1}.$$

For a constant α, $x = \{\alpha, \alpha, \alpha, \ldots\}$ has z-transform $X = \dfrac{\alpha z}{z-1}$.

exponential growth

In the series

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + \cdots$$

replace z by $2/z$:

$$\frac{1}{1-2z^{-1}} = 1 + 2z^{-1} + 4z^{-2} + 8z^{-3} + 16z^{-4} + 32z^{-5} + 64z^{-6} + \cdots$$

So the z-transform of $x = \{1, 2, 4, 8, 16, 32, 64, \ldots\}$ is

$$X = \frac{1}{1-2z^{-1}} = \frac{z}{z-2}.$$

Observe: the essence of the signal, the growth rate is the root of $z - 2$.

exponential decay

In the series

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + \cdots$$

replace z by $1/(2z)$:

$$\frac{1}{1-\frac{1}{2}z^{-1}} = 1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2} + \frac{1}{8}z^{-3} + \frac{1}{16}z^{-4} + \frac{1}{32}z^{-5} + \frac{1}{64}z^{-6} + \cdots$$

So the z-transform of $x = \left\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \ldots\right\}$ is

$$X = \frac{1}{1-\frac{1}{2}z^{-1}} = \frac{z}{z-\frac{1}{2}}.$$

The essence of the signal, the decay rate is the root of $z - 1/2$.

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the z-transform of a delayed signal

$x = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, \ldots\}$ has z-transform $X = \sum_{k=0}^{\infty} x_k z^{-k}$.

Let y be the signal x with one time unit delay:

$$y = \{0, x_0, x_1, x_2, x_3, x_4, x_5, x_6, \ldots\}.$$

What is the z-transform Y of y?

$$X = \sum_{k=0}^{\infty} x_k z^{-k} = x_0 + \frac{x_1}{z} + \frac{x_2}{z^2} + \frac{x_3}{z^3} + \frac{x_4}{z^4} + \frac{x_5}{z^5} + \frac{x_6}{z^6} + \cdots$$

Apply the definition of the z-transform:

$$Y = \frac{x_0}{z} + \frac{x_1}{z^2} + \frac{x_2}{z^3} + \frac{x_3}{z^4} + \frac{x_4}{z^5} + \frac{x_5}{z^6} + \frac{x_6}{z^7} + \cdots = z^{-1} X.$$

taking weighted averages

Let x be a signal with z-transform X.

Let us make a new signal y, with a weighted average.

$$y_k = \alpha x_k + \beta x_{k-1}, \quad \alpha + \beta = 1, \quad x_{-1} = 0.$$

What is the z-transform Y of y?

$$\begin{aligned} Y &= \sum_{k=0}^{\infty} y_k z^{-k} = \sum_{k=0}^{\infty} \left(\alpha x_k + \beta x_{k-1} \right) z^{-k} \\ &= \sum_{k=0}^{\infty} \alpha x_k z^{-k} + \sum_{k=0}^{\infty} \beta x_{k-1} z^{-k} \\ &= \alpha X + \beta z^{-1} X = (\alpha + \beta z^{-1}) X \end{aligned}$$

a first exercise

Exercise 1:
Find the z-transform of 1, 2, 3, 1, 2, 3, 1, 2, 3, . . .

linear combinations

Exercise 2:

Show that the z-transform of a linear combination of two signals is the linear combination of the two z-transforms of the signals.

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a recursion for our motivating problem

A train leaves the depot empty.

At each station, on average,

- 40 people mount
- while 20% of the riders dismount.

We can define a recursion relation for this problem.

1. Let x_n be the number of riders at the n-th station.
   $x_0 = 0$, as the train leaves the depot empty.

2. As 20% dismount, 80% or 4 out of five remain, forty are added:

$$x_n = \left(\frac{4}{5} \right) x_{n-1} + 40, \quad n = 1, 2, \ldots$$

solving recursions via the z-transform

Given n fixed constants $\alpha_1, \alpha_2, \ldots, \alpha_n$, a linear recursion is

$$x_k = \alpha_1 x_{k-1} + \alpha_2 x_{k-2} + \cdots + \alpha_n x_{k-n},$$

where n initial values $x_{-1}, x_{-2}, \ldots, x_{-n}$ are known.

Solving a linear recursion proceeds in three steps:

1. Transform the recursion to the frequency domain.
2. Solve the transformed recursion.
3. Transform the solution back to the time domain.

applied to our motivating problem

We derived the linear recursion:

$$x_n = \left(\frac{4}{5}\right) x_{n-1} + 40, \quad n = 1, 2, \ldots$$

The z-transform is

$$X = \left(\frac{4}{5}\right) z^{-1} X + 40 \frac{z}{z-1}.$$

We solve to obtain X as an expression in z:

$$\left(1 - \frac{4}{5} z^{-1}\right) X = 40 \frac{z}{z-1}$$

$$X = \left(\frac{40z}{z-1}\right) \left(\frac{5z}{5z-4}\right) = 200 \frac{z^2}{(z-1)(5z-4)}$$

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partial fraction decomposition

Consider a quotient of two polynomials $\dfrac{p(x)}{q(x)}$.

- Let $q(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$ with mutually distinct roots: $\alpha_i \neq \alpha_j$, for all $i \neq j$.
- Let $p(x)$ be any polynomial of degree $\deg(p) < n$.

Then the *partial fraction decomposition* of the quotient is

$$\frac{p(x)}{q(x)} = \sum_{i=1}^{n} \frac{p(\alpha_i)}{q'(\alpha_i)} \left(\frac{1}{x - \alpha_i} \right).$$

Observe: $\alpha_i \neq \alpha_j$, for all $i \neq j$ implies $q'(\alpha_i) \neq 0$.

proof via Lagrange interpolation

The Lagrange form of the polynomial interpolating $p(x)$ through $\alpha_1, \alpha_2, \ldots, \alpha_n$ is

$$p(x) = \sum_{i=1}^{n} p(\alpha_i) \prod_{\substack{j=1 \\ j \neq i}}^{n} \left(\frac{x - \alpha_j}{\alpha_i - \alpha_j} \right).$$

Because

- the interpolation problem has a unique solution, and
- $\deg(p) < n$,

the Lagrange form of the interpolating polynomial equals $p(x)$.

rewriting the Lagrange interpolating polynomial

$$p(x) = \sum_{i=1}^{n} p(\alpha_i) \prod_{\substack{j=1 \\ j \neq i}}^{n} \left(\frac{x - \alpha_j}{\alpha_i - \alpha_j} \right) \left(\frac{x - \alpha_i}{x - \alpha_i} \right)$$

$$= \sum_{i=1}^{n} \left(\frac{p(\alpha_i)}{x - \alpha_i} \right) \underbrace{\prod_{j=1}^{n} \left(x - \alpha_j \right)}_{q(x)} \prod_{\substack{j=1 \\ j \neq i}}^{n} \left(\frac{1}{\alpha_i - \alpha_j} \right)$$

$$\frac{p(x)}{q(x)} = \sum_{i=1}^{n} \frac{p(\alpha_i)}{q'(\alpha_i)} \left(\frac{1}{x - \alpha_i} \right)$$

as $q'(x) = \sum_{i=1}^{n} \prod_{\substack{j=1 \\ j \neq i}}^{n} (x - \alpha_j)$.

Q.E.D.

returning to our motivating problem

For the z-transform, we found:

$$X = 200 \frac{z^2}{(z-1)(5z-4)} = 40z \frac{z}{(z-1)(z-4/5)}.$$

Apply

$$\frac{p(x)}{q(x)} = \sum_{i=1}^{n} \frac{p(\alpha_i)}{q'(\alpha_i)} \left(\frac{1}{x - \alpha_i} \right).$$

to

$$\begin{aligned} \frac{z}{(z-1)(z-4/5)} &= \frac{1}{q'(1)} \left(\frac{1}{z-1} \right) + \frac{4/5}{q'(4/5)} \left(\frac{1}{z-4/5} \right) \\ &= 5 \left(\frac{1}{z-1} \right) - 4 \left(\frac{1}{z-4/5} \right) \end{aligned}$$

the solution

Combining

$$X = 40z \frac{z}{(z-1)(z-4/5)}$$

and

$$\frac{z}{(z-1)(z-4/5)} = 5\left(\frac{1}{z-1}\right) - 4\left(\frac{1}{z-4/5}\right)$$

gives

$$X = 200\left(\frac{z}{z-1}\right) - 160\left(\frac{z}{z-4/5}\right) = 200\left(\frac{z}{z-1} - \frac{(4/5)z}{z-4/5}\right).$$

Observe that we find 200 as an upper bound.

the solution in the time domain

We compute the inverse of $X = 200 \left(\underbrace{\frac{z}{z-1}}_{X_1} - \underbrace{\frac{(4/5)z}{z-4/5}}_{X_2} \right)$.

1. Let x_1 be the inverse of X_1: $x_1 = \{1, 1, 1, \ldots\}$.
2. Let x_2 be the inverse of X_2: $x_2 = \frac{4}{5} \left\{ 1, \frac{4}{5}, \left(\frac{4}{5}\right)^2, \ldots \right\}$.

Now we make a linear combination:

$$x = \left\{ 1 - \frac{4}{5}, 1 - \left(\frac{4}{5}\right)^2, 1 - \left(\frac{4}{5}\right)^3, \ldots \right\}, \text{ or } x_k = 200 \left(1 - \left(\frac{4}{5}\right)^k \right),$$

for $k = 1, 2, \ldots$, using $x_0 = 0$.

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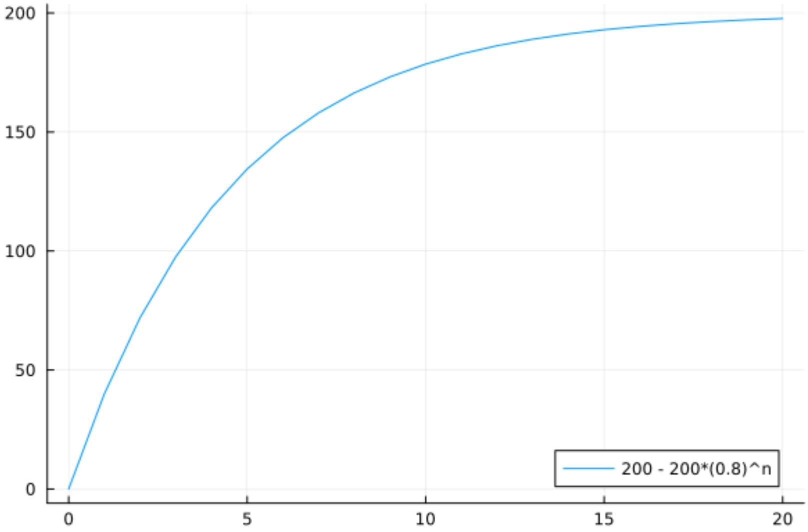
using `sympy.rsolve`

```
julia> using SymPy

julia> n = Sym("n")
n

julia> x = SymFunction("x")
x

julia> eq = Eq(x(n), (4/5)*x(n-1) + 40)
x(n) = 0.8x(n - 1) + 40

julia> init = Dict(x(0) => 0)
Dict{Sym, Int64} with 1 entry:
  x(0) => 0

julia> sympy.rsolve(eq, x(n), init)
                 n
200.0 - 200.0 0.8
```

a plot of the solution



our motivating problem solved

A train leaves the depot empty.

At each station, on average,

- 40 people mount,
- while 20% of the riders dismount.

Suppose we are consultants for a train company, who asks us:

Will there ever be a train long enough

to accommodate the 40 people mounting at each station?

The solution to our motivating problem is $200 - 200(0.8)^n$, so a train with 200 seats will accommodate the 40 people mounting at each station.

running a simulation

Exercise 3:

Set up a Monte Carlo simulation for our motivating problem:

1. At each stop, each rider on board of the train flips a loaded coin so on average one out of five on board dismount.
2. Run the simulation for 20 stops, each simulation results in a vector of 20 integers, counting the number of riders on board.
3. Run the simulation sufficiently many times (e.g.: 1,000) and make a plot of the average counts for each stop. Compare your plot with the plot of the solution of the recursion.

one more exercise

Exercise 4:
Consider the compound interest rate i.
$x(n)$ is the balance after compounding n times.

1. Apply the z-transform method to solve

$$x(n+1) = (1+i)x(n), \quad x_{-1} = 1/(1+i).$$

2. Use `sympy.rsolve` to solve the recursion.

one last exercise

Exercise 5:

Compute the partial fraction decomposition of $\dfrac{1}{(z-1)(z-2)(z-3)}$.

You may use computer algebra for this computation.

bibliography

We covered the start of Chapter 3 of

- Charles R. MacCluer **Industrial Mathematics. Modeling in Industry, Science, and Government**, Prentice Hall, 2000.

Z-transforms were introduced in

- J.R. Ragazzini and L.A. Zadeh:
  The Analysis of Sampled-Data Systems. *Transactions of the American Institute of Electrical Engineers, Part II: Applications and Industry.* 71 (5): 225-234, 1952.